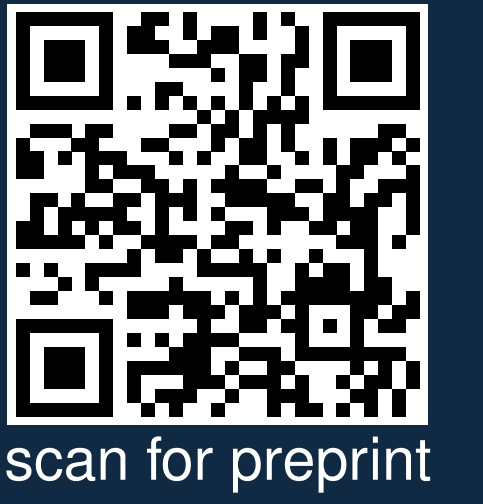


Dawn and Twilight Time in Quantum Tunneling

When does exponential decay actually begin and end?

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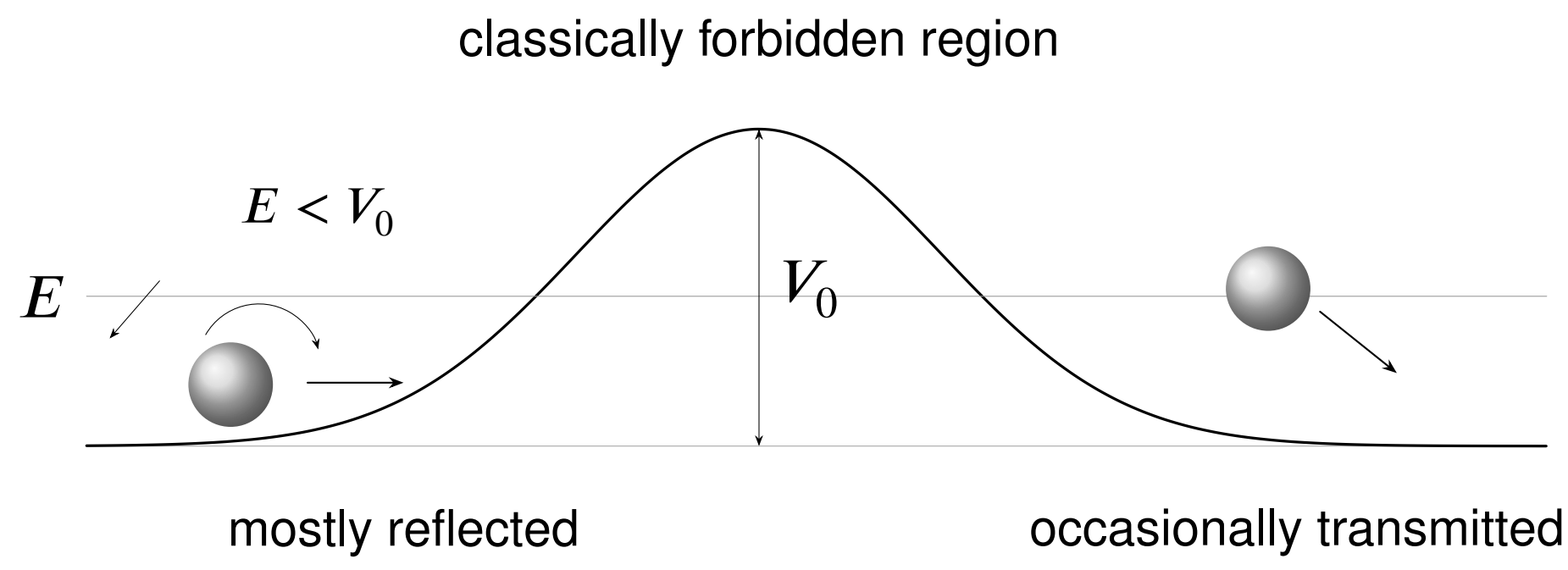
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arXiv:2512.14809



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Main point: Exponential decay is not the whole story; it is a middle-time window.

1. Scientific question



Metastable decay is often compressed into

$$P_L(t) \sim e^{-\Gamma t}.$$

But exact real-time evolution is not purely exponential:

early transient $\longrightarrow$ exponential plateau $\longrightarrow$ power-law tail

What is the actual time window where the rate Γ is meaningful?

2. Real-time definition

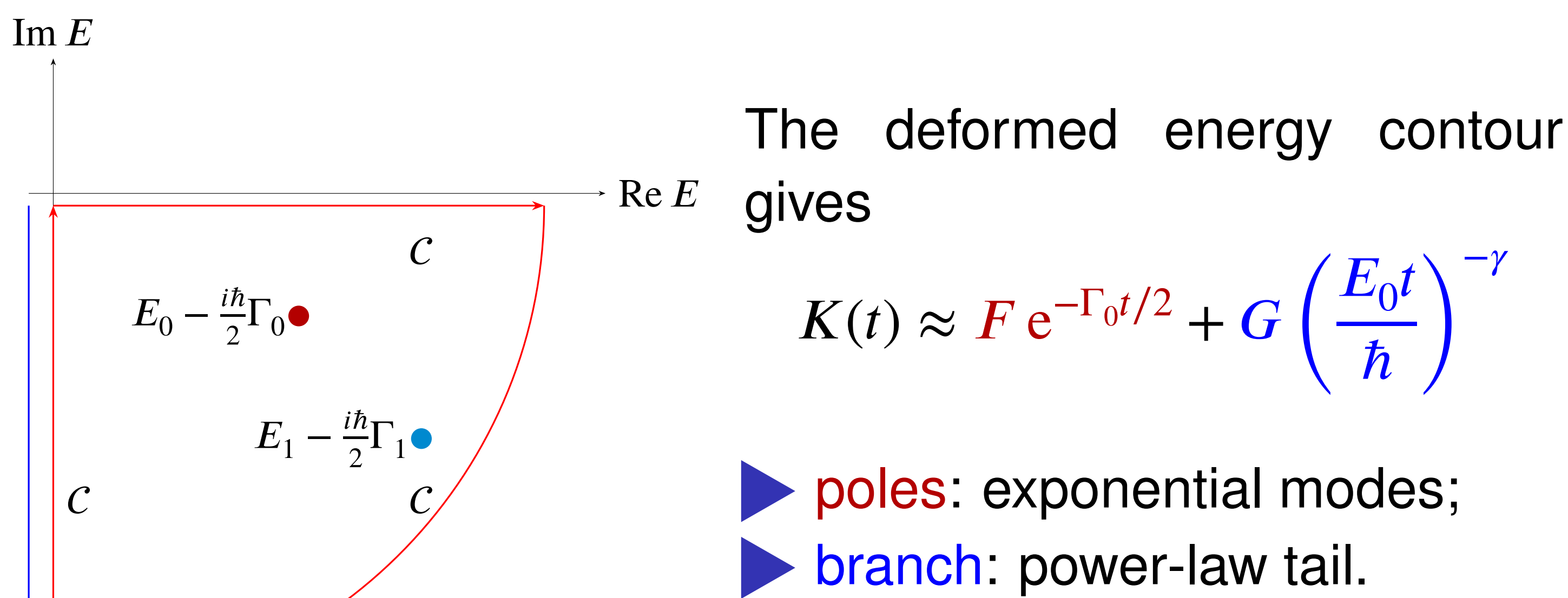
Use probability flux out of the metastable region:

$$\Gamma(t) = -\frac{1}{P_L(t)} \frac{dP_L(t)}{dt}, \quad P_L(t) = \int_L dx |\Psi(x, t)|^2.$$

The wavefunction evolves through the kernel

$$\Psi(x, t) = \int_L dy K(t, x; 0, y) \Psi(y, 0).$$

3. Spectral decomposition

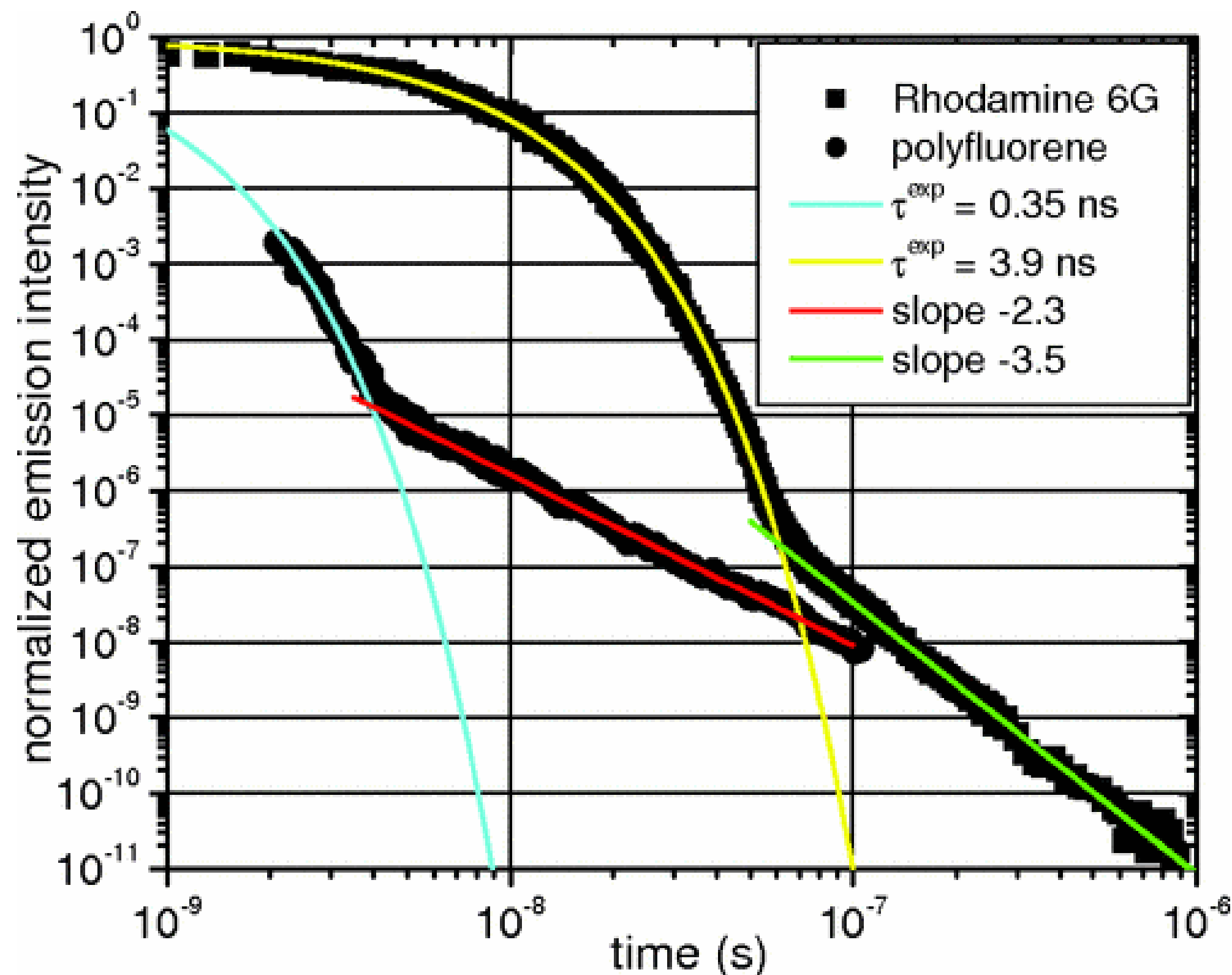


4. What is calculated

- **Dawn time $t_\ominus$** : when the leading pole starts controlling the decay.
- **Twilight time $t_\omin�$** : when the power-law tail overtakes the exponential.
- **Physical check**: for a thick square barrier, $\Gamma T = T_{\text{trans}}$.

5. Experimental hint: long-time tails

Long-time non-exponential decay is not just a formal possibility; related power-law tails have been observed in tunneling experiments with cold atoms.



Example of exponential-to-power-law crossover; Rothe et al., PRL 96 (2006).

6. Main result: exponential window

$$t_\ominus \leq t \leq t_\omin�$$

The standard exponential law is justified only inside this window.

7. The two timescales

Twilight: the late crossing of pole and power-law tail,

$$\left(\frac{E_0 t_\omin�}{\hbar}\right)^{-2\gamma} = \left|\frac{F}{G}\right|^2 e^{-\Gamma_0 t_\omin�},$$

which gives

$$t_\omin� = -\frac{2\gamma}{\Gamma_0} W_{-1} \left[-\frac{\Gamma_0 \hbar}{2\gamma E_0} \left|\frac{G}{F}\right|^{1/\gamma} \right]$$

Dawn: the first time at which the leading pole beats both higher poles and the early branch contribution,

$$t_\ominus = \max \left\{ \max_n \left[\frac{1}{\Gamma_n - \Gamma_0} \ln \left(\frac{\tilde{F}(E_n)}{\tilde{F}(E_0)} \right) \right], -\frac{2\gamma}{\Gamma_0} W_0 \left[-\frac{\Gamma_0 \hbar}{2\gamma E_0} \left|\frac{G}{F}\right|^{1/\gamma} \right] \right\}$$

Dawn depends on the initial wave packet; twilight is mostly spectral.

8. Analytic checks: thick-barrier twilight

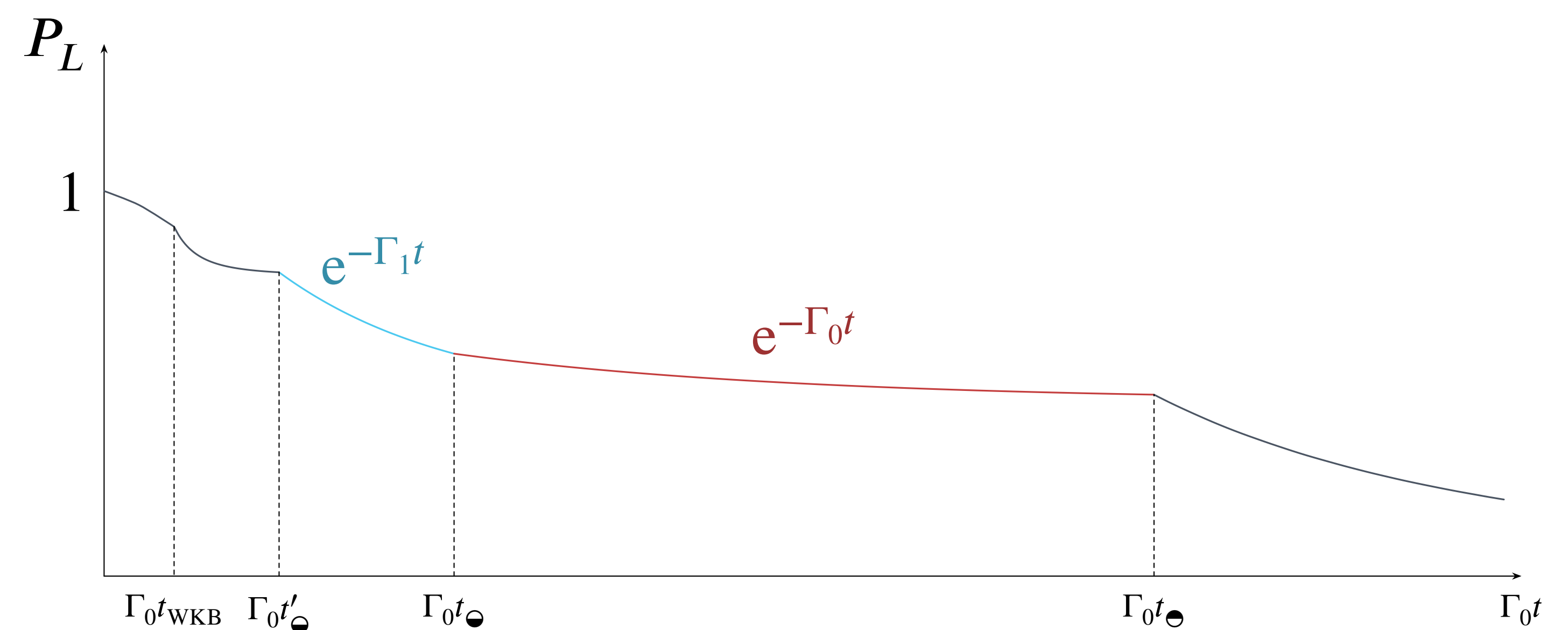
Model	γ	thick-barrier $t_\omin�$
square	3/2	$\frac{6\kappa(b-a)}{\Gamma_0 \hbar}$
modified square	2	$\frac{8\kappa(b-a)}{\Gamma_0 \hbar}$
Pöschl–Teller	1/2	$\frac{2\pi\kappa}{\Gamma_0 \alpha \hbar}$

9. Example: square barrier

For the initial state

$$\Psi_0(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi\mu x}{a}\right),$$

we get the qualitative decay history of P_L :



The usual exponential law appears only after early transients and before the late tail.

Conclusion

1. Γ is the slope of a middle-time plateau, not the full decay law.
2. The real-time kernel separates pole physics from the power-law tail.
3. The exponential approximation is justified only for $t_\ominus \leq t \leq t_\omin�$.

Γ gives the slope; $t_\ominus$ and $t_\omin�$ give the domain of validity.